

Roll No.

UCE-501

B. TECH. (EE) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2022

DISASTER MANAGEMENT

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) How important are vulnerability and risk assessment for pre-disaster management ? As an administrator, what are key areas that you would focus in a disaster management ? (CO1 and CO5)

(b) Explain causes and effects of the following Natural Disasters :

(CO1 and CO5)

(i) Earthquake

(ii) Flood

(iii) Landslide

(iv) Tsunami

P. T. O.

- (c) Explain the various phases of disaster management in detail.
(CO1 and CO5)
2. (a) Explain how disaster preparedness is the first step in any disaster management process.
(CO3 and CO6)
- (b) The novel coronavirus outbreak has become a worldwide disaster. How far is it correct to say such frequent viral outbreaks are manmade disasters ?
(CO3 and CO6)
- (c) Mention the different phases of Disaster Management Cycle and how the phases are related to each other.
(CO3 and CO6)
3. (a) What are landslides ? List out *five* major causes of landslides in India. Suggest risk reduction measures for landslide mitigation.
(CO3 and CO4)
- (b) Describe Vulnerability, Risk, Capacity and Early warning.
(CO3 and CO4)
- (c) Describe the different institutions at the Central, State and District level involved in disaster management.
(CO3 and CO4)
4. (a) How is biodiversity loss associated with infrastructure development ? How does environmental impact assessment help in reconciling the developmental needs ?
(CO2)
- (b) How developmental activities promote the chances of disaster ? Explain any one disaster caused by developmental activities.
(CO2)
- (c) Explain the importance of disaster management course for engineers.
(CO2)

(3)

5. (a) What do you mean by structural and non-structural mitigation of Disaster Management ? (CO5 and CO6)
- (b) Explain the role of GIS and MIS methods of response. (CO5 and CO6)
- (c) Himalaya may be entering a phase of irreversible decline because of its losses to its ecology. Discuss in the light of frequent disasters in the region. (CO5 and CO6)

Roll No.

TEE-501

B. TECH. (EE) (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

POWER SYSTEM-I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Compare the volume of the conductor material required in three-phase three wire systems and single-phase two wire DC system with mid-point earth. Assume same line loss, same power and same line length.

(CO1/CO2)

- (b) A 2-wire feeder carries a constant current of 250 A throughout the year. The portion of capital cost which is proportional to area of X-section is ₹ 5 per kg of copper conductor. The interest and depreciation total 10% per annum and the cost of energy is 5 P per kWh.

P. T. O.

Find the most economical area of X-section of the conductor. Given that the density of copper is 8.93 gm/cm^3 and its specific resistance is $1.73 \times 10^{-8} \Omega\text{-cm}$. (CO1/CO2)

- (c) A single-phase ac system supplies a load of 200 kW and if this system is converted to 3-phase, 3-wire system by running a third similar conductor, calculate the 3-phase load that can now be supplied if the voltage between the conductor is the same. Assume the power factor and transmission efficiency to be in the two cases. (CO1/CO2)

2. (a) What do you mean by string efficiency ? Discuss the various methods of improving the string efficiency of suspension type insulator. (CO3)

- (b) A 3-phase, 220 kV, 50 Hz transmission line consists of 1.5 cm radius conductor spaced 2 metres apart in equilateral triangular formation. If the temperature is 40°C and atmospheric pressure is 76 cm, calculate the corona loss per km of the line. Take $m_0 = 0.85$. (CO3)

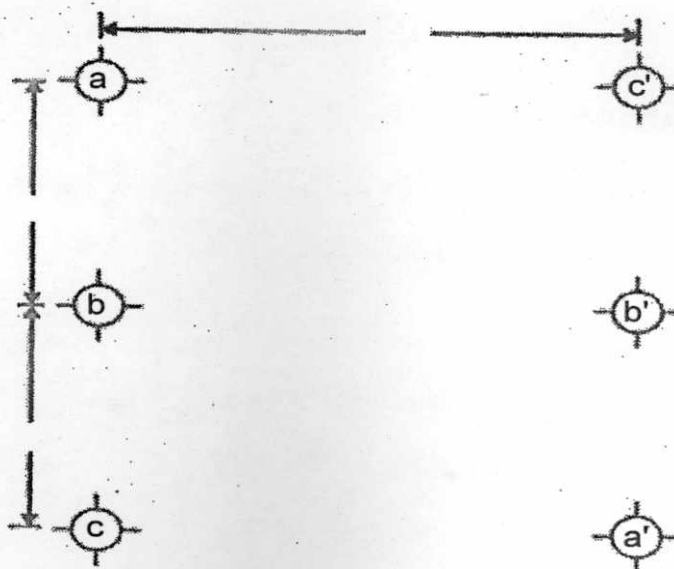
- (c) An insulator string consists of three units, each having safe working voltage of 15 kV. The ratio of self-capacitance to shunt capacitance of each unit is 8 : 1. Find the maximum safe working voltage of the string. Also find the string efficiency. (CO3)

3. (a) Derive the approximate of mechanical sag for unequal supports. (CO6)

- (b) What is Corona ? What are the factors which affect corona ? (CO6)

- (c) The towers of height 30 m and 90 respectively support a transmission line conductor at water crossing. The horizontal distance between the towers is 500 m. If the tension in the conductor is 1600 kg, find the minimum clearance of the conductor and water and clearance mid-way between the supports. Weight of conductor is 1.5 kg/m. Bases of the towers can be considered to be at water level. (CO6)

4. (a) Derive the expression of internal and external flux linkage of conductor carrying current. (CO5)
- (b) Deduce the expression for line to neutral capacitance for 3-phase 3-wire ac system when the conductor is symmetrically spaced. (CO5)
- (c) The spacing between double circuit lines are shown in the figure. If radius of each conductor is 1.3 cm, then find the transmission line per phase and per kilometer. (CO5)



5. (a) Derive the expression for insulation resistance and capacitance of single core underground cable. (CO4, CO6)
- (b) What do you mean by grading of underground cable ? Explain inter-sheath grading of underground cable. (CO4, CO6)
- (c) A single core lead sheathed cable has a conductor diameter of 3 cm; the diameter of the cable being 9 cm. The cable is graded by using two dielectrics of relative permittivity 5 and 4 respectively with corresponding safe working stresses of 30 kV/cm and 20 kV/cm. Calculate the radial thickness of each insulation and the safe working voltage of the cable. (CO4, CO6)

Roll No.

TEE-502

**B. TECH. (ELECTRICAL ENGINEERING)
(FIFTH SEMESTER)**

END SEMESTER EXAMINATION, Dec., 2022

CONTROL SYSTEMS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

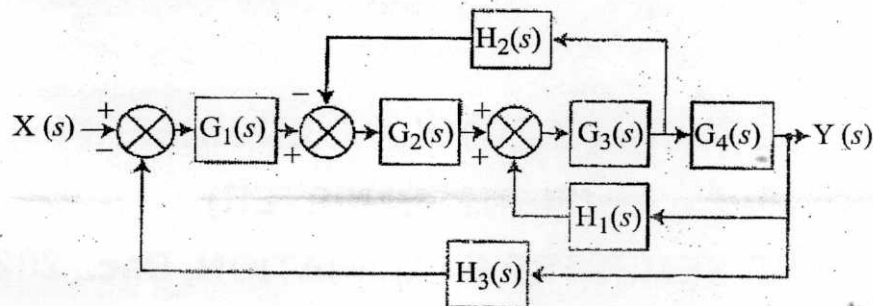
(iv) Each sub-question carries 10 marks.

1. (a) Explain the block diagram of control systems. What are the various categories of control systems ? Explain each of them in brief. (CO1)
- (b) What is the objective of using Laplace Transformation in Control Systems ? Find the Laplace transform of the following test signals used in control systems : (CO1)
 - (i) Unit Impulse
 - (ii) Unit Ramp
 - (iii) Unit Step

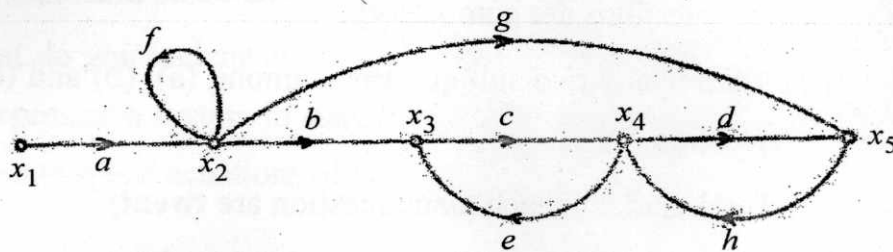
P. T. O.

(c) Distinguish between open and closed loop control systems. (CO1)

2. (a) Construct the over-all transfer function of depicted block diagram by reduction method : (CO2)



- (b) Describe the use of the Masson's gain formula in control systems. Apply Mason's gain formula to obtain the overall transmittance of the following signal flow graph : (CO2)



- (c) Obtain the position, velocity and acceleration error coefficients for a system given by : (CO2)

$$G(s) = \frac{100(s+2)(s+40)}{s^3(s^2+4s+200)}$$

3. (a) Examine the time response of a first order system, as given below and is subjected to unit step input (i. e. $R(s) = 1/s$). (CO3)

$$\frac{C(s)}{R(s)} = \frac{2}{(s+1)}$$

(b) Describe the following as associated to the time response of system :

(CO3)

- Rise time (t_r)
- Peak time (t_p)
- Peak overshoot (M_p)
- Settling time (t_s)

(c) What do you understand by stability ? Describe absolute and relative stability. List the various techniques used to study the stability of a system. (CO4)

4. (a) What do you understand by stability ? Describe absolute and relative stability. List the various techniques used to study the stability of a system. (CO4)

(b) Investigate the stability of the system using Routh Criterion whose characteristic equation is given by : (CO4)

$$5s^5 + s^4 + 2s^3 + 3s^2 + 2s + 1 = 0.$$

(c) The open loop transfer function of a system is given by : (CO4 & 5)

$$G(s)H(s) = \frac{1}{s(1 + 0.5s)(1 + 0.08s)}$$

Determine :

- (i) Gain margin
- (ii) Phase margin and
- (iii) Closed loop stability

P. T. O.

5. (a) Consider a unity feedback control system with the following open loop transfer function : (CO5)

$$G(s)H(s) = \frac{K}{s(s+1)(s+3)}$$

Plot the root loci of the system.

Also, Discuss the procedure to design the phase lag compensation using Root Locus.

- (b) Explain the following terms as associated to the control systems : (CO5 & 6)
- (i) Controllability and Observability
 - (ii) Gain margin and Phase margin
 - (iii) Gain and phase crossover frequencies
- (c) What do you understand by the state of a system ? Write the equations to represent a system in state space. Also, draw the block diagram using the state space equations of system. (CO6)

Roll No.

TOE-502

B. TECH. (EE) (FIFTH SEMESTER) END SEMESTER EXAMINATION, Dec., 2022

COMPUTER BASED NUMERICAL AND STATISTICAL TECHNIQUES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) Discuss Truncation Error. Find the absolute, relative and percentage errors if x is rounded off to three decimal digits. Given $x = 0.005998$.

(CO1)

- (b) In a ΔABC , $a = 6$ cm, $c = 15$ cm, $\angle B = 90^\circ$. Find the possible error in the computed value of A if the errors in measurements of a and c are 1 mm and 2 mm respectively.

(CO1)

- (c) The error in the measurement of the area of a circle is not allowed to exceed 0.1%. How accurately should the diameter be measured ?

(CO1)

P. T. O.

2. (a) Starting with $x = 0.12$, solve $x = 0.21 \sin(0.5 + x)$ by using iteration method. (CO2)
- (b) Apply Newton's formula to find values $(30)^{1/5}$. (CO2)
- (c) Find the root of the equation $\tan x + \tanh x = 0$ which lies in the interval $(1.6, 3.0)$ correct to four significant digits using method of false position. (CO2)
3. (a) Define Interpolation. What are the assumptions made for interpolation ? (CO3)
- (b) Find the cubic polynomial which takes the following values : (CO3)
 $y(0) = 1, y(1) = 0, y(2) = 1$ and $y(3) = 10$
Hence or otherwise obtain $y(4)$.
- (c) Using Lagrange's interpolation formula, find $y(10)$ from following table : (CO3)

x	y
5	12
6	13
9	14
11	16

4. (a) Discuss why numerical differentiation should be avoided if an alternative exists. What are the various methods available for numerical integration ? (CO4)

(3)

- (b) The table below gives the result of an observation. θ is the observed temperature in degrees centigrade of a vessel of cooling water, t is the time in minutes from the beginning of observations : (CO4)

t	θ
1	85.3
3	74.5
5	67.0
7	60.5
9	54.3

Find the approximate rate of cooling at $t = 3$.

- (c) Find, from the following table, the area bounded by the curve and the x -axis from $x = 7.47$ to $x = 7.52$: (CO4)

x	$f(x)$
7.47	1.93
7.48	1.95
7.49	1.98
7.50	2.01
7.51	2.03
7.52	2.06

5. (a) Numerical methods of solution of O. D. E. yield the solution in which one of the two forms, what are those two forms ? Discuss the drawbacks of Taylor's method. (CO5, CO6)

- (b) Solve : (CO5, CO6)

$$\frac{dy}{dx} = 1 + xy \text{ with } x_0 = 2, y_0 = 0$$

using Picard's method.

- (c) Give $\frac{dy}{dx} = y - x$, $y(0) = 2$. Find $y(0.1)$ correct to four decimal places by using fourth order Runge-Kutta method. (CO5, CO6)

Roll No.

TEE-503

B. TECH. (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022

MICROPROCESSORS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) (i) How is the physical address generated in 8086 microprocessor ?

(ii) How is the memory segmentation done in 8086 microprocessor ?

(CO1 & CO3)

(b) Explain the register organization of 8086 microprocessor.

(CO1 & CO3)

(c) List the timing and control signal controlled by timing and control unit of 8085 microprocessor. Explain any *five*.

(CO1 & CO3)

P. T. O.

2. (a) Explain the addressing modes of 8086 microprocessor for sequential instructions along with examples. (CO2)
- (b) Explain the following instructions of 8085 : (CO2)
- (i) DAA
 - (ii) XCHG
 - (iii) RET
 - (iv) DAD rp
 - (v) LDAX rp
- (c) Write an assembly language program of 8085 microprocessor to find out largest number in an array of 8 bit numbers starting from memory location 3001 H. The size of array is 10 H. Store result at 2050 H. (CO2)
3. (a) Explain the following instructions of 8086 microprocessor : (CO3)
- (i) MUL
 - (ii) DAS
 - (iii) NEG
 - (iv) DIV
 - (v) ROL
- (b) Draw and explain the architecture of 8086 microprocessor. (CO3)
- (c) Write an assembly language program to transfer a block of 8 bit numbers from 1000 H : 5000 H to 2000 H : 6000 H. The size of string is FF H. (CO3)

(3)

4. (a) Define the handshaking signals. Describe the various data transfer schemes. (CO4)
- (b) Draw and explain the block diagram of Programmable peripheral interfacing IC 8255. (CO4)
- (c) Draw the control word format of PPII C 8255 for both I/O mode and Bit set reset mode. (CO4)
5. (a) Draw and explain the block diagram of DMA Controller IC 8257. (CO5, CO6)
- (b) Compare Memory mapped I/O with Peripheral mapped I/O. (CO5, CO6)
- (c) Draw the timing diagram of MVI A 00H. (CO5, CO6)

Roll No.

TEE-504

**B. TECH. (FIFTH SEMESTER)
END SEMESTER EXAMINATION, Dec., 2022**

ELECTRICAL MACHINE DESIGN

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are twenty.

(iv) Each question carries 10 marks.

1. (a) Derive the output equation of a single phase core type transformer.

(CO1, CO2)

(b) Define specific magnetic loading. Explain the factors need to be considered for choice of specific magnetic loading in a dc machine.

(CO1, CO2)

(c) The average flux density in the air gap and specific electric loading are respectively 0.57T and 30000 ampere-conductors per metre. Calculate suitable dimensions of armature core to lead to a square pole face. Assume that full load armature drop is 3% of the rated voltage and the field current is 1% of rated full load current. Ratio pole arc to pole pitch is 0.67.

(CO1, CO2)

P. T. O.

2. (a) Explain in steps show to separate D and L for a DC machine.
(CO2, CO3)
- (b) Derive the output power of an induction motor incorporating specific electric and magnetic loading.
(CO2, CO3)
- (c) Determine the main dimensions of a 500 kVA, 50 Hz 3-phase alternator to run at 375 rpm. The average air gap flux density is 0.55 Wb/mm^2 , the specific electric loading is 25000 A/m. The peripheral speed should not exceed 35 m/sec.
(CO2, CO3)
3. (a) Explain the procedure for separation of D and L from D^2L product while designing induction motors.
(CO4, CO6)
- (b) Differentiate between square pole face and rectangular pole face.
(CO4, CO6)
- (c) A design is required for a 50kW, 4 pole, 600 rpm, and 220 V dc shunt generator. Determine the number of poles, armature diameter and core length for the preliminary design of a 500 kW, 400 V, 600 rpm, dc shunt generator assuming an average flux density in the air gap of 0.7 T and specific electric loading of 38400 ampere-conductors per metre. Assume core length/pole arc = 1.1. Apply suitable checks. (CO4, CO6)
4. (a) List the factors to be considered for the choice of specific electric loading in synchronous machines.
(CO4, CO5)
- (b) What are the guiding factors that decide the selection of ampere conductors per metre of an induction machine ?
(CO4, CO5)

(3)

- (c) A 3-phase, 16 pole synchronous generator has a star connected winding with 144 slots and 10 conductors per slot. The flux per pole is 0.04 Wb and speed is 375 rpm. Find the frequency and phase induced emf. The turns/phase may be assumed to be series connected. (CO4, CO5)
5. (a) What is meant by specific electric and specific magnetic loading in a dc machine ? From first principles, obtain the expressions for specific magnetic loading and specific electric loading. (CO6, CO5)
- (b) Derive the output equation of DC machine. (CO6, CO5)
- (c) Calculate fundamental, third and fifth harmonic breadth factors for a stator with 36 slots wound for three phase 4 pole. (CO6, CO5)